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How alcoholic beverages like beer and wine is made?

The conversion of pyruvate to ethanol takes place in malted barley and grapes through fermentation. Yeasts carryout this process under anaerobic conditions and this conversion increases ethanol concentration. If the concentration increases, it's toxic effect kills yeast cells and the left out is called beer and wine respectively.

14.9 Pentose Phosphate Pathway (Phospho Gluconate Pathway)

During respiration breakdown of glucose in cytosol occurs both by glycolysis (about 2/3) as well as by oxidative pentose phosphate pathway (about 1/3). Pentose phosphate pathway was described by **Warburg**, **Dickens** and **Lipmann** (1938). Hence, it is also called **Warburg-Dickens-Lipmann pathway**. It takes place in cytoplasm of mature plant cells. It is an alternate way for breakdown of glucose (Figure 14.15).

It is also known as **Hexose monophosphate shunt (HMP Shunt)** or **Direct Oxidative Pathway**. It consists of two phases, oxidative phase and non-oxidative phase. The oxidative events convert six molecules of six carbon Glucose-6-phosphate to 6 molecules of five carbon sugar Ribulose-5 phosphate with loss of $6CO_2$ molecules and generation of 12 NADPH + H^+ (not NADH). The remaining reactions known as **non-oxidative pathway**, convert Ribulose-5-phosphate molecules to various intermediates such as Ribose-5-phosphate(5C),Xylulose-5-phosphate(5C), Glyceraldehyde-3-phosphate(3C), Sedoheptulose-7-Phosphate(7C), and Erythrose-4-phosphate(4C). Finally, five molecules of glucose-6-phosphate is regenerated (Figure 14.16). The overall reaction is:

6 x Glucose-6-Phosphate + 12$NADP^+$ + 6H_2O

↓

5 x Glucose-6-Phosphate + 6CO_2 + Pi + 12NADPH +12H^+

The net result of complete oxidation of one glucose-6-phosphate yield $6CO_2$ and 12NADPH + H^+. The oxidative pentose phosphate pathway is controlled by glucose-6-phosphate dehydrogenase enzyme which is inhibited by high ratio of NADPH to $NADP^+$.

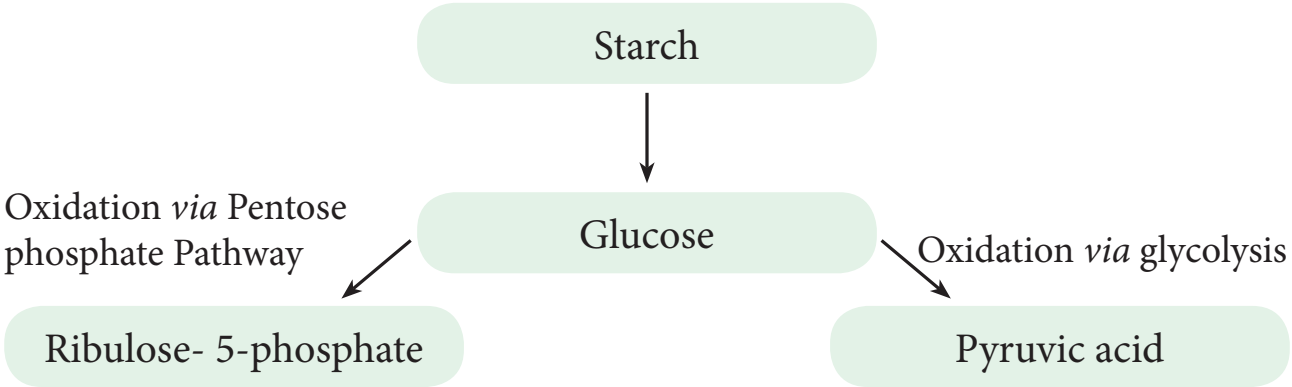


Figure 14.15: Fate of Glucose in HMP shunt and Glycolysis